A2 BIOLOGY

5A: Photosynthesis

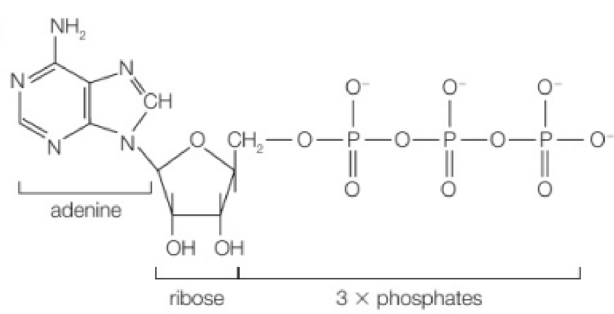
Energy is needed for life, and categories have been made based on how the organisms get their energy:

- **Autotrophic** - Produces organic compounds from inorganic compounds. Autotrophic nutrition can be further divided into two categories.
    - **Photosynthetic** - Use Light to bind water and carbon dioxide to form organic compounds
    - **Chemosynthetic** - Use oxidation reactions of inorganic compounds to form organic compounds.

- **Heterotrophic** - Obtain energy by eating other plants or animals that have eaten plants. They use the products of photosynthesis indirectly to form their organic molecules.

ATP

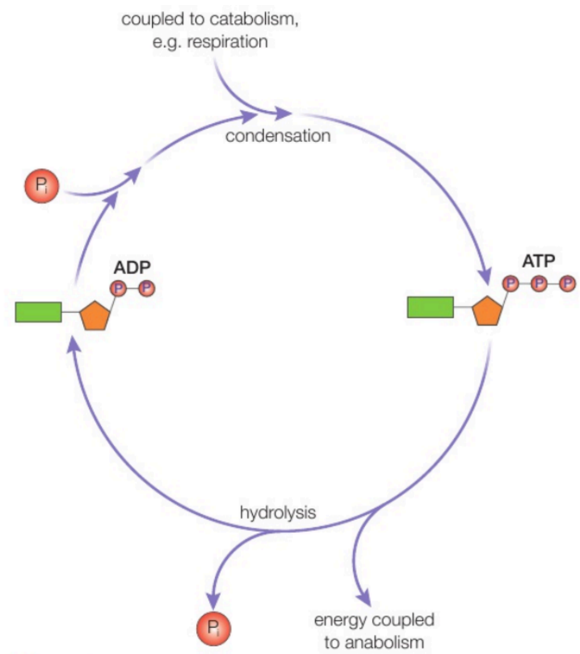
ATP stands for Adenosine Triphosphate. It is found in all living organisms in the same form. Energy is released when **ATPase** breaks down ATP into ADP (Adenosine Diphosphate). A free phosphate along with energy is released in a hydrolysis reaction.



1 mole of ATP can produce up to 34kJ of energy.

The energy for the reverse reaction is produced by catabolic reactions(redox reactions).

The electron transport chain is a series of protein complexes embedded in the inner mitochondrial membrane that transfers electrons and generates ATP in cellular respiration. It creates a proton gradient that drives ATP synthesis through chemiosmosis. It is essential for energy production in cells.



CHLOROPLASTS AND CHLOROPHYLL

Photosynthesis is the conversion of light energy to chemical potential energy. The process involves two main stages, which can be simplified into one equation.

$$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$$

The green pigment chlorophyll and sunlight are needed for this reaction to occur. This reaction occurs mainly in the chloroplasts of the palisade cells.

Chlorophyll

There are two types of chlorophyll

- Chlorophyll A - Dark green / Blue-green
- Chlorophyll B - Light green / Yellow-green

Chlorophyll is a light-absorbing group of molecules that converts light energy into chemical energy.

Along with chlorophyll, other pigments are also needed to maximise the efficiency.

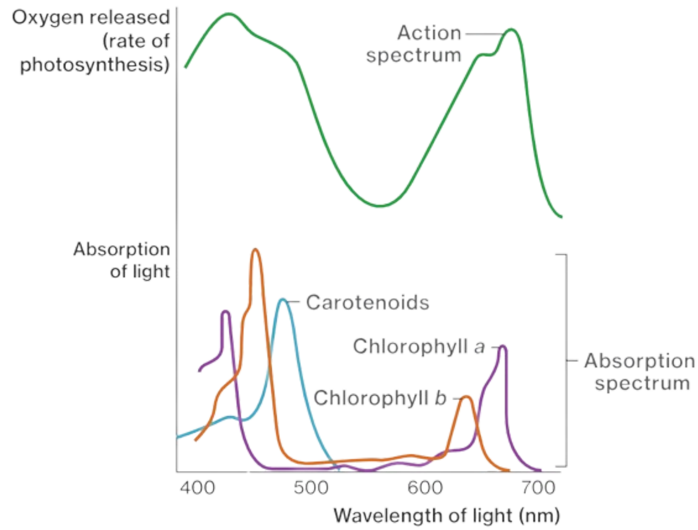
These accessory pigments are Carotenoids and Xanthophylls that filter the wavelengths of light supplied only to allow the needed wavelengths (in the PSI) 680 nm and 700 nm. As a result, more energy can be harnessed compared to only one pigment.

Absorption Spectra

Absorption spectra are essential for understanding the different wavelengths of light absorbed by various pigments.

Chlorophyll a, the primary pigment, absorbs light most efficiently at 430 nm (blue) and 662 nm (red), while chlorophyll b absorbs best at 453 nm (blue) and 642 nm (red).

These specific wavelengths are vital for the energy conversion processes.



Action Spectra

In photosynthesis, the action spectra of pigments refer to the efficiency with which different pigments absorb light at various wavelengths. Here is a short introduction to the action spectra of pigments during photosynthesis:

- **Chlorophyll**: The main pigment involved in photosynthesis is chlorophyll, which absorbs light most efficiently in the blue and red regions of the spectrum. Chlorophyll reflects green light, and as a result, plants appear green to our eyes.
- **Accessory pigments**: Plants also have other pigments like carotenoids and Xanthophylls that are accessory pigments that filter light.
- **Action spectra**: The action spectra of pigments show the relative effectiveness of different wavelengths of light in driving photosynthesis. It helps us understand which wavelengths of light are most efficiently absorbed by the pigments.
- **Peak absorption**: Different pigments have peak absorption at specific wavelengths. For example, chlorophyll a absorbs light most efficiently at

wavelengths around 430 nm and 662 nm, while chlorophyll b absorbs best at around 453nm and 642 nm.

Chloroplasts

Chloroplast is a double membrane-bounded organelle that is present in plant cells.

Structure

- Double Membrane
  - Outer Membrane
  - Inner Membrane - Invaginated (folded in and out)
- Lamella System
  - Grana + inter-grana lamella structures joined together.
- Stroma
  - Ground material
  - storage/ **suspend organelles**/molecule exchange
  - **WHERE LIGHT-INDEPENDENT REACTIONS OCCUR**
- Grana
  - Stacks of Thylakoids
- Thylakoids
  - The single disk-like structure where photosynthesis occurs
  - **WHERE LIGHT-DEPENDENT REACTIONS OCCUR**
- Lipid droplets
  - Energy store
  - Appear as dark/black spots in an electron micrograph
- Starch granules
  - Energy store
  - Appear as light/white spots in an electron micrograph larger than lipid droplets.

- Ribosomes
    - Transcription of proteins
- DNA strand
    - Carry genetic information for replication and growth of the chromosomes and the Synthesis of proteins
    - In charge of chloroplast replication

Types of reactions in the chloroplasts

Light dependant reactions- happen in the thylakoids

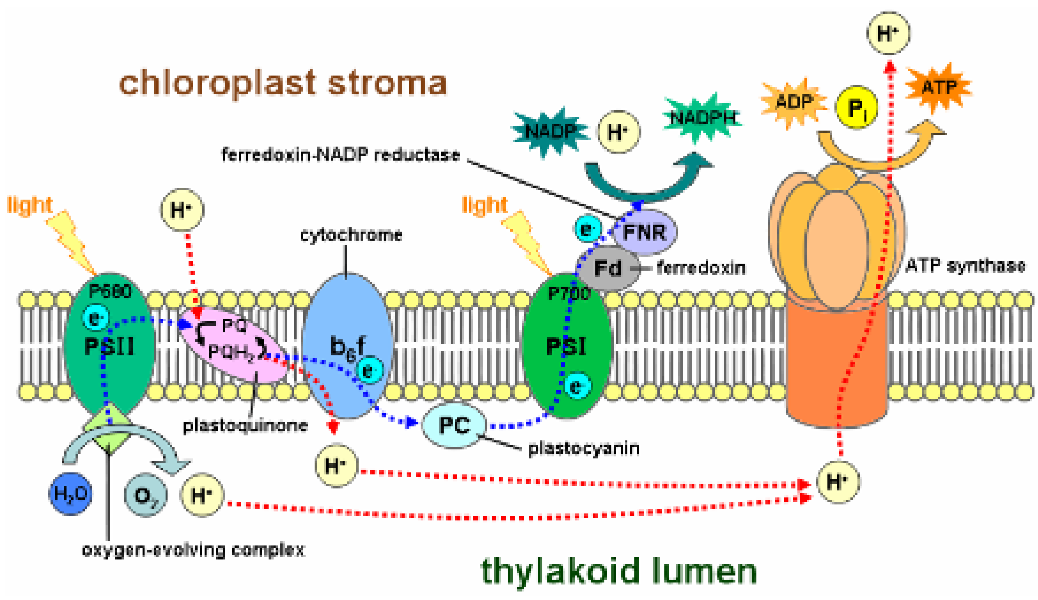
Light-independent reactions- happen in the stroma

Light dependant reactions convert ADP to ATP and NADP (Nicotinamide Adenine Dinucleotide Phosphate) to NADPH

Thylakoid

THYLAKOID MEMBRANE

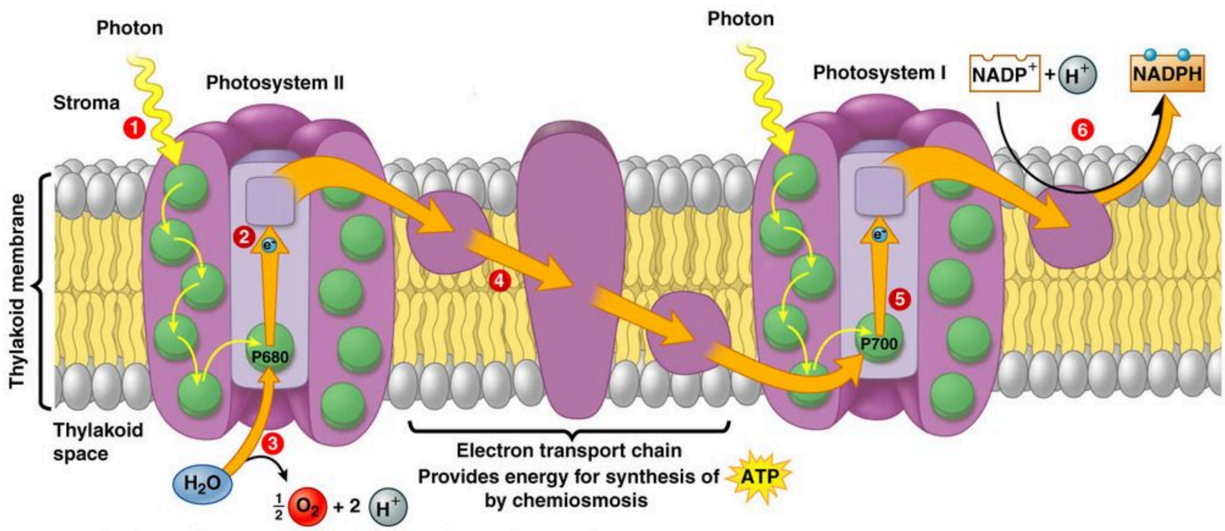
A phospholipid bilayer with intrinsic proteins different from those in the cell membrane



Examples of intrinsic proteins in the thylakoid membrane

- PSI (photosystem 1)
- PSII (photosystem 2)
- Cytochrome
- ATP synthase
- Plastoquinone
- Plastocyanin

PHOTOSYSTEMS



- There are 2 photosystem types namely photosystem one and photosystem two (PSI and PSII)
- These are cup-like structures that have molecules of different stains like xanthophyll and carotenoids (that absorb wavelengths other than 680 nm and 700 nm) that filter the photons of light that are received by it and supply light of one of two discrete wavelengths to the main reaction centre found at the apex centre of the cone or cup.
  - 700 nm - for the chloroplasts in the PSI
  - 680 nm - for the chloroplasts in the PSII
- The centre chlorophyll can be **only p680 or p700.**
- PSI has a special enzyme called a photolytic enzyme that breaks down water.

CYTOCHROME

The cytochrome b6f complex is a protein that helps transport electrons and pumps H^+ ions into the thylakoid lumen.

PLASTOQUINONE AND PLASTOCYANIN

Plastoquinone and plastocyanin are electron acceptors that help move electrons along the electron transport chain.

ATP SYNTHASE

Produces ATP from ADP by using the electron from the electron transport chain.

THYLAKOID LUMEN

This is where photophosphorylation occurs

protons(Hydrogen Nuclei) are pumped into the lumen.

How Palisade cells are adapted for their function

- Cylindrical/elongated shape - Large Surface area for absorption of light
- Lots of chloroplasts - maximising photosynthesizing ability and increasing rate of photosynthesis
- Large Permanent Vacuole - pushes chloroplasts to the periphery of the cell and also helps store material like sucrose
- Internal Cyclosis - keeps chloroplast constantly in motion maximising exposure to sunlight
- Near upper epidermis - cells closer to the surface maximising sunlight exposure

PROCESS OF PHOTOSYNTHESIS

The process of photosynthesis occurs in two main steps

- Light Dependent Reactions
- Light Independent Reactions

Each step has its requirements and it takes place simultaneously. One process can't take place without the other.

Light Dependent Reactions

These processes occur at the thylakoids of the chloroplast.

The main processes that take place are:

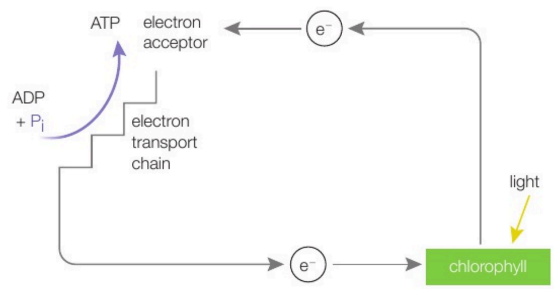
- Photolysis of water
- Produce ATP

Light-dependent reactions take place in two different ways, both of which are used.

- Cyclic Photophosphorylation
- Noncyclic Photophosphorylation

Cyclic Photophosphorylation

This involves only the PSI and the electrons excited by light passing through the electron acceptor. It then passes through the electron transport chain, producing ATP. This electron then passes on back to the photosystem.



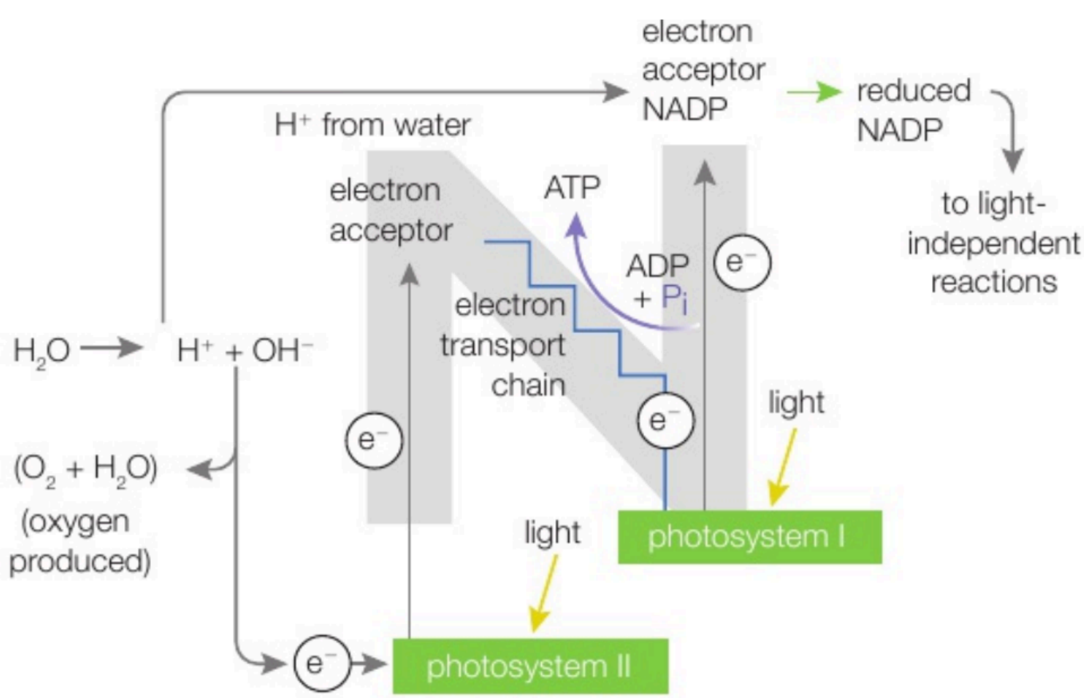
Noncyclic Photophosphorylation

The pigments' electrons get excited as light reaches the photosystem II, moving from one bond to another until they reach the chlorophyll in the middle. At this point, the pigments leave the photosystem onto the plastoquinone (electron acceptor I). This causes the chlorophyll to gain a Positive charge. For the chlorophyll to function again, it should be neutral. To do this, the photolysis of water is used to release electrons which neutralise the charge of the chlorophyll.

$$2\ H_2O \rightarrow 4\ e^- + 4\ H^+ + O_2$$

The electrons from the chlorophyll then proceed across the electron transport chain through the cytochrome and plastocyanin proteins. While the electron passes through cytochrome, ADP and P_i (inorganic phosphate) are converted into ATP as a coupled reaction.

At the same time, when light reaches the photosystem II, 4 electrons are released to the ferredoxin attached to ferredoxin NADP reductase, which converts NADP+ to reduced NADP, which moves on to the light-independent reactions. The ionised chlorophyll in the PSII is neutralised by the electrons from PSI



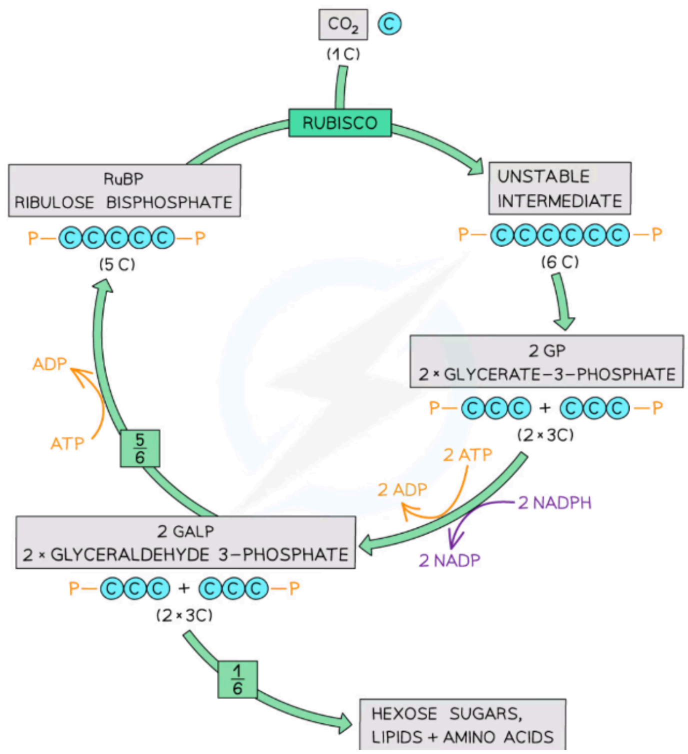
Light Independent Reactions

These processes take place at the stroma and use the products of light-dependent reactions.

This stage consists of repeating reactions known as the Calvin cycle.

Calvin Cycle

The Calvin Cycle is a series of chemical reactions during photosynthesis.



1. Carbon Fixation:

RuBisCO catalyses the fixation of atmospheric carbon dioxide by attaching it to a five-carbon sugar molecule called ribulose bisphosphate (RuBP).

This reaction results in an unstable six-carbon compound that splits into two molecules of glycerate-3-phosphate (GP).

2. Reduction Phase:

Each molecule of GP is then phosphorylated by ATP and NADPH transfers high-energy electrons and protons to it, converting it into glyceraldehyde-3-phosphate (GALP). Some GALP molecules are used to regenerate RuBP, while others are utilised to build glucose and other carbohydrates.

3. **Regeneration of RuBP:**

The remaining GALP molecules are rearranged and phosphorylated to regenerate RuBP, the molecule needed for the carbon fixation step to continue the cycle. This regeneration process requires ATP.

4. **Overall Outcome:**

$6CO_2$ (6 carbons)+ 6RuBP (30 carbons) → 12GP (36 carbons)

12GP (36 carbons) + 12 NADPH + 12 ATP → 12 GALP (36 carbons)

12 GALP(36 carbons)→ Glucose (6 Carbons) + 6 RuBP (30 carbons)

USES OF THE PRODUCTS OF PHOTOSYNTHESIS

GALP - Glyceraldehyde - 3 Phosphate

1. Directly For respiration
2. Conversion to Glucose - can be converted to:
    a. Disaccharides
    b. Polysaccharides
    c. Cellulose
    d. Amino acids (add an amine group)
    e. Fatty acids and glycerol which make Lipids

Oxygen

For respiration in all living organisms.

LIMITING FACTORS IN PHOTOSYNTHESIS

The rate of photosynthesis depends on many different factors. A decrease in any of these factors reduces the rate of photosynthesis.

1. Light

Without light, the light dependent reactions cannot occur, reducing the amount of reduced NADP and ATP produced.

☆ **Light Intensity** and **Wavelength** are limiting factors

2. Carbon Dioxide

Carbon Dioxide is a main reactant during the calvin cycle. Without carbon dioxide, Glucose cannot be made.

☆ **Concentration** of Carbon dioxide is a Limiting Factor.

3. Temperature

Since enzymes play a large role in both the light dependent and light independent reactions, a change in temperature will affect the activity of the enzymes, where large fluctuations can permanently stop reactions in the plant.

CORE PRACTICAL 10

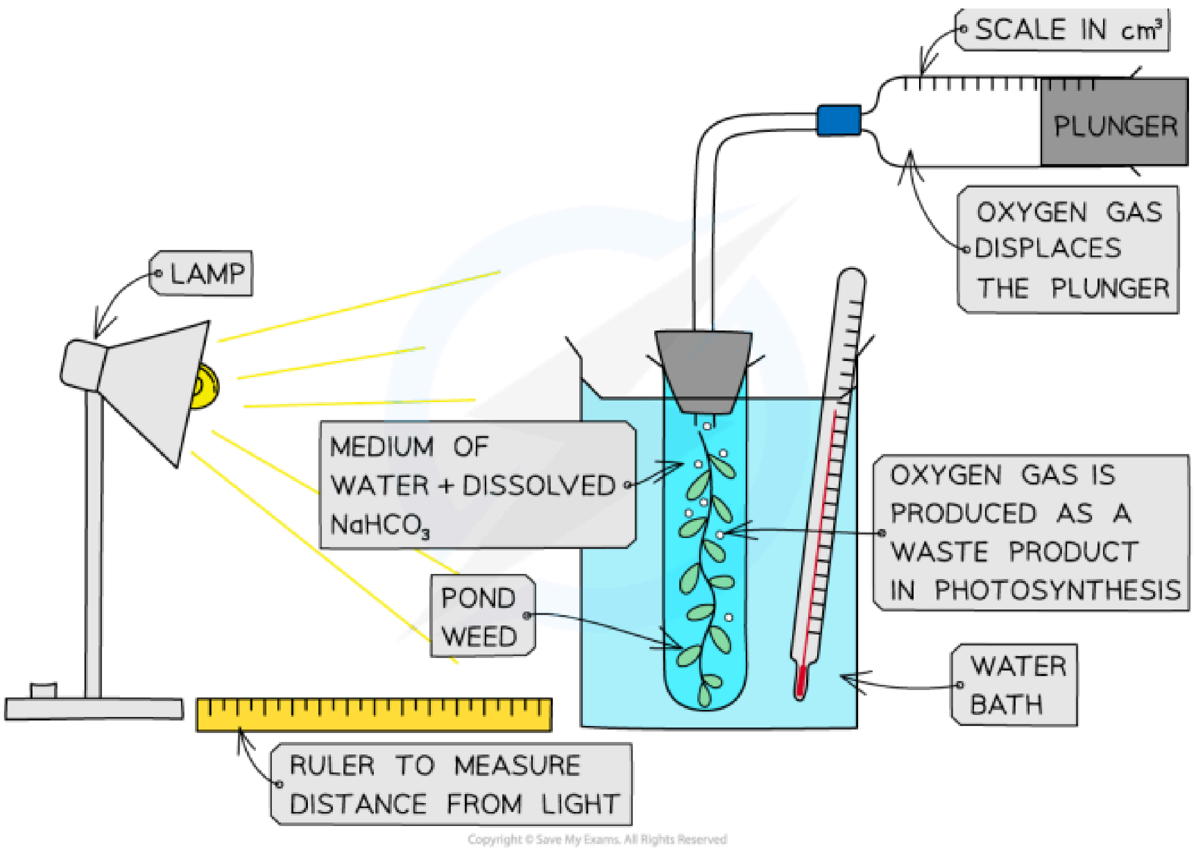
Apparatus

- Distilled water
- Boiling tube
- Beaker
- Lamp
- Aquatic plant, algae or algal beads
- Metre ruler
- Sodium hydrogen carbonate solution
- Thermometer
- Boiling tube bung and delivery tube
- Measuring syringe

Method

1. Optional: ensure the water is well aerated before use by bubbling air through it.
   a. This will ensure that any oxygen gas given by the plant during the investigation forms bubbles instead of dissolving
2. Ensure the plant has been well illuminated before use.
   a. This will ensure that the plant contains all the enzymes required for photosynthesis and that any changes of rate are due to changes in light intensity
3. Set up the apparatus in a darkened room with a 10 cm distance between the lamp and the boiling tube.
   a. This means that no external light sources will affect the rate of photosynthesis
4. Cut the stem of the pondweed cleanly just before placing it into the boiling tube.
   a. This enables bubbles of oxygen to form from the cut plant stem.
5. Submerge the sample of pondweed in sodium hydrogen carbonate solution.
   a. This ensures that the pondweed has a controlled supply of carbon dioxide

6. Measure The Volume Of Gas collected in the gas syringe over a set period of time, e.g.1 minute; repeat this measurement 3 times
7. Change The Independent variable by moving the light source10 cm further away from the plant and repeat step 6
8. Repeat step 7 several more times, moving the lamp10 cm further away from the plant each time
9. Record the results in a table and plot a graph of volume of oxygen produced per minute against the distance from the lamp.



Results

The closer the lamp,the higher the light intensity, therefore the volume of oxygen produced should increase as the light intensity is increased. At a point the volume of oxygen produced will stop increasing even if the light is moved closer. This is the point at which light intensity stops being the limiting factor and, e.g. temperature or concentration of carbon dioxide becomes limiting.

5B: Ecology

WHAT IS ECOLOGY

Ecology is the study of the environment or relationship among living organisms and their physical environment.

Terms

- **Individual**: a single member of a species.
- **Population**: Numbers of individuals of a species.
- **Community**: Number of different populations in an area.
- **Ecosystems**: the relationships and interactions between all organisms within an area and their physical environment.
- **Biome**: the community of plants and animals that occur naturally in an area, often sharing common characteristics specific to the area.
- **Biosphere**: the region of life where life can exist and grow.
- **Niche**: The role played by an organism in an ecosystem.
- **Abiotic Factors**: The non-living factors in an ecosystem.
- **Biotic Factors**: The living factors in an ecosystem.

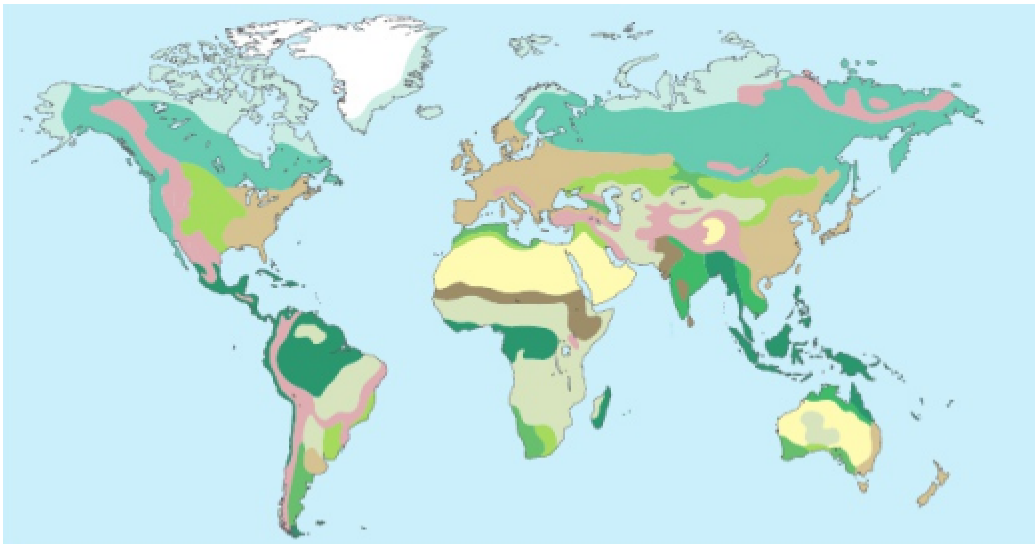
Ecosystems

An ecosystem is an environment including all the living organisms interacting within it, the cycling of nutrients and the physical and chemical environment in which the organisms are living.

An ecosystem can consist of many different habitats and the connections between different habitats and their organisms.

Biomes

The biosphere is the largest ecosystem on Earth. Its vastness means it cannot be studied as one, therefore we study smaller parts of the Earth.



COLOUR	NAME OF BIOME	DESCRIPTION OF BIOME	LEVEL OF BIODIVERSITY
	tropical rainforest	high humidity (rain all year), warm and plenty of sunlight	very high
	tropical seasonal forest	drier than tropical rainforest, warm, sunny	high
	savannah	dry tropical grassland	medium
	tropical woodland	wetter than savannah, grassland with bushes and trees	more than savannah
	desert	very little rainfall, often extremes of temperature between day and night	very low
	temperate grassland	warm dry temperate areas, e.g. prairies, steppes and pampas	medium
	temperate shrublands	hot dry summers and cool wet winters	medium
	temperate forests	warm moist regions, including deciduous and conifers	less than tropical rainforest
	taiga	evergreen forests in cold subarctic and subalpine regions	low
	tundra	very cold, arctic and high mountain regions	very low
	high mountain	very cold, high altitude	very low
	polar ice	very cold, little available water	very low

HOW ECOSYSTEMS EVOLVE

All the biomes existing now have been made over millions of years. All the biomes were formed from bare rock to what it is now by the help of succession of multiple generations of plants and animals.

Succession

Primary Succession

Primary succession starts off on an inorganic surface like bare rocks.

Stage 1 - Colonisation

The first organisms to appear(pioneer species) are algae, mosses and fungi. They grow and penetrate the rock by excreting acids or penetrate cracks using hyphae. As the weather changes each season, they die out and grow back. The dead algae and mosses form humus which help maintain moisture and nutrients, supporting the growth of new plants

Stage 2 - Intermediate Stage I

After a few years, the rock breaks down to sand and the layer of humus thickens and grass starts growing. This is boosted by the nutrients added to the soil from dead plants and humus from the previous stage

Stage 3 - Intermediate Stage II

The death of grasses add to the soil and the soil is now able to support the growth of larger shrubs and ferns.

Stage 4 - Climax Stage

Taller trees start growing and develop the soil enough for a more diverse range of plants to grow. Eventually a more dominant species takes root and the Climax community is achieved.

Secondary Succession

Secondary Succession starts from existing bare soil left from forest fires, floods or human activity.

Since there is soil present, the growth of plants is much faster than primary succession. The climax community reached may be different from the original climax community. It depends on the first colonisers.

EFFECT OF ABIOTIC FACTORS ON POPULATIONS

Abiotic factors refer to non-living components of an ecosystem that can influence the growth, development, and survival of populations.

1. Temperature:

- Temperature plays a crucial role in determining the distribution of species.
- Extreme temperatures can impact the enzymes in charge of metabolic rates of organisms, affecting their growth and reproduction.
- Organisms have specific temperature ranges within which they can survive and thrive.

2. Water:

- Water availability is essential for the survival of all living organisms.
- Drought conditions can lead to dehydration, affecting population sizes.
- Aquatic organisms are particularly sensitive to changes in water quality and quantity.

3. Light:

- Light intensity and photoperiod (length of day) influence processes like photosynthesis, which is vital for the energy flow within ecosystems.
- Plants rely on light for photosynthesis, and changes in light availability can impact plant growth and, consequently, the populations that depend on them.

4. Soil:

- Soil composition, pH, and nutrient levels can affect the types of plants that grow in an area.
- The availability of nutrients in the soil can impact plant productivity, which in turn influences the populations of herbivores and predators in the ecosystem.

5. Wind:

- Wind can affect the dispersal of seeds and spores, influencing plant populations.
- Strong winds can also cause physical damage to plants, leading to a decline in population numbers.

6. Altitude:

- Altitude influences factors like temperature, air pressure, and oxygen levels, which can limit the distribution of certain species.
- Organisms adapted to high altitudes have physiological adaptations to cope with lower oxygen levels.

7. Pollution:

- Pollution from human activities can have detrimental effects on populations by causing habitat degradation and contamination of air, water, and soil.
- Pollutants can disrupt ecosystems and lead to declines in population numbers or even extinctions.

Understanding how abiotic factors impact populations is essential for conservation efforts and ecosystem management, as changes in these factors can have far-reaching consequences on the biodiversity and stability of ecosystems.

Mnemonic: *LETCOW*

Light, Edaphic(soil), Temperature, Currents, Oxygen, Water

EFFECT OF BIOTIC FACTORS ON POPULATIONS

Biotic factors are living components of an ecosystem that can have significant impacts on populations of organisms.

1. Competition:

Interspecific and intraspecific competition for resources such as food, water, and shelter can influence population size and distribution. It can lead to adaptations and niche differentiation among species.

2. Predation:

Predators play a crucial role in regulating prey populations. Predation can control prey population sizes and prevent overgrazing or overpopulation of certain species.

3. Parasitism:

Parasites can negatively impact host populations by weakening individuals and reducing their reproductive success. This relationship can influence population dynamics and overall ecosystem health.

4. Disease:

Pathogens and diseases can spread through populations, affecting individuals' health and potentially leading to population declines or fluctuations. Disease outbreaks can have cascading effects on an ecosystem.

5. Mutualism:

Mutualistic relationships, where both species benefit, can enhance population growth and survival. For example, pollinators and plants rely on each other for reproduction and can support each other's populations.

6. Symbiosis:

Symbiotic relationships, including mutualism, commensalism, and parasitism, can impact population dynamics by influencing species interactions and the overall balance of the ecosystem.

Understanding how biotic factors interact with populations is essential for studying ecology and ecosystem dynamics. By examining these relationships, we can gain insights into the delicate balance of nature and the factors that shape populations within ecosystems.

ECOSYSTEM INTERACTIONS AND THE IMPORTANCE OF THE NICHE

Ecosystem interactions refer to the various relationships and connections between organisms within an ecosystem. These interactions can take many forms, including competition, predation, mutualism, and parasitism. Understanding ecosystem interactions is crucial for ecologists as they help to explain how energy and nutrients flow through an ecosystem.

The Importance of the Niche:

- **Definition**: The niche of an organism refers to its role or function within an ecosystem, including how it obtains its resources, interacts with other species, and its impact on the environment.
- **Role in Ecosystem**: Each species occupies a unique niche that contributes to the overall balance and functioning of the ecosystem.
- **Resource Partitioning**: Species within an ecosystem occupy different niches to reduce competition for resources. This leads to better coexistence and biodiversity.
- **Species Interactions**: Understanding the niche of a species helps predict its interactions with other species, such as competition, predation, or mutualism.
- **Ecosystem Stability**: Niches help maintain the stability of an ecosystem by preventing one species from dominating and causing imbalance.

Ecosystem interactions and the importance of the niche are fundamental concepts in biology that help us understand how different species coexist, compete, and contribute to the overall functioning and stability of ecosystems.

Competition

Competition occurs when two or more organisms in an ecosystem require the same resources that are in limited supply. These resources can include food, water, shelter, sunlight, or mates.

Types of Competition:

Intraspecific Competition:

This type of competition occurs between individuals of the same species. For example, two lions fighting over a kill.

Interspecific Competition:

This type of competition occurs between individuals of different species. For example, lions and hyenas competing for the same prey.

Effects of Competition:

Resource Partitioning:

Competition can lead to resource partitioning, where species evolve to occupy different niches to reduce competition.

Competitive Exclusion:

Intense competition can lead to competitive exclusion, where one species outcompetes another, leading to the elimination of the weaker species from that habitat.

Evolutionary Adaptations:

Competition can drive evolutionary adaptations in species, leading to changes in behaviour, morphology, or physiology to reduce competition.

Example: In a forest ecosystem, different species of birds may compete for nesting sites. Some species may nest at different heights in trees or use different types of materials to build their nests to reduce direct competition.

INVESTIGATING ABUNDANCE AND DISTRIBUTION OF ORGANISMS

Abundance

Refers to the number of individuals of a particular species in a given area at a specific time. It is essential to quantify the abundance of organisms to understand their population size and how it changes over time.

Distribution

Refers to the spatial arrangement of individuals of a species in a particular area. Understanding the distribution of organisms helps in identifying patterns and factors influencing their habitat preferences.

Sampling Techniques

Various methods are used to sample populations for study, such as quadrat sampling, transect sampling and belt transect sampling. These techniques help in obtaining representative data on abundance and distribution.

Quadrat Sampling

Definition : Quadrat sampling estimates abundance and distribution of organisms.

Purpose: Estimating population density, distribution patterns, and species diversity.

Design: Use random sampling, various quadrat types.

Size: Choose appropriate quadrat size for organisms and research questions.

Data: Collect species counts, cover percentages, or relevant measurements.

Analysis: Calculate population densities, species richness, and use statistical tests.

Errors: Understand sources of sampling errors like edge effects and observer bias.

Line Transect Video

Definition: Line transects involve counting and recording organisms along a straight line.

Method: Use standardised surveys along the line to gather data on distribution and abundance.

Design: Ensure a straight, well-marked transect line for accurate data collection.

Data Collection: Record organism numbers along the line to estimate population densities.

Analysis: Use data to assess species richness, diversity, and ecosystem dynamics.

Importance: Crucial for monitoring biodiversity and understanding ecosystem changes.

Belt Transect Video

Definition: Belt transects involve sampling a continuous strip of habitat to study the distribution of organisms.

Method: Sampling along a designated strip of habitat.

Data Collection: Counting and recording organisms within the strip.

Analysis: Provides insights into species distribution and abundance.

Importance: Useful for studying habitat preferences and biodiversity patterns.

Measuring Abiotic Factors

ABIOTIC FACTOR	METHOD OF MEASUREMENT
temperature of air, soil, water, etc.	glass or digital thermometers; thermistors (temperature probes, particularly useful for measuring water temperature)
light intensity	light meters
pH	chemical pH tests based on pH indicators; pH meters
wind speed	anemometers
slope of an incline	clinometers
oxygen levels (in water)	digital oxygen probe

STATISTICS AND ECOLOGY

Null Hypothesis

A null hypothesis is a statement that suggests there is no significant difference or relationship between groups or variables being studied. It serves as a basis for statistical hypothesis testing. Researchers aim to reject the null hypothesis in favour of an alternative hypothesis based on the data collected during an experiment or study. The null hypothesis assumes there is no effect or relationship until evidence proves otherwise.

Spearman's Rank Correlation Coefficient

Spearman's rank correlation coefficient is a statistical measure that assesses the strength and direction of association between two ranked variables.

Here is how Spearman's rank correlation coefficient method works:

- Ranking: The method involves ranking the data for each variable from lowest to highest.
- Calculating Differences: The differences between the ranks for each pair of observations are calculated.
- Squaring Differences: The squared differences are calculated for each pair of ranks.
- Summing Ranks: The sum of the squared differences is calculated.
- Calculating the Coefficient: The Spearman's rank correlation coefficient is calculated using a formula that takes into account the number of observations and the sum of the squared differences.

Rank Set A	Rank Set B	d	d^2
1	3	2	4
2	3	1	1
3.5	1	2.5	6.25
3.5	3	0.5	0.25
5	6	1	1
6	5	1	1

$$\sum d^2 = 4 + 1 + 6.25 + 0.25 + 1 + 1 = 13.5$$

From (1-1)

$$r_s = 1 - \frac{6\sum d^2}{n(n^2-1)}$$

$$r_s = 1 - \frac{6 \cdot 13.5}{6(6^2-1)}$$

$$r_s = 0.61428571$$

$$r = \frac{\sum(x-\overline{x})(y-\overline{y})}{\sqrt{\sum(x-\overline{x})^2 \sum(y-\overline{y})^2}}$$

Key points about Spearman's rank correlation coefficient method:

- It is used when the data does not follow a normal distribution.
- It assesses the monotonic relationship between variables (whether the variables tend to increase or decrease together, but not necessarily at a constant rate).
- The coefficient ranges from -1 to 1, where:-
  - 1: Perfect negative correlation-
  - 0: No correlation-
  - 1: Perfect positive correlation

Spearman's rank correlation coefficient is particularly useful when dealing with ordinal data or when the assumptions of Pearson's correlation coefficient are not met.

Student's t-test

The Student's t-test is a statistical test used to determine whether the difference between the responses of two groups is statistically significant. It is commonly used to compare the means of two groups and assess if the difference observed could have occurred by chance. The test statistic in a t-test follows a Student's t-distribution, which is a probability distribution that accounts for the variability in small sample sizes. This test helps researchers make inferences about population means based on sample data.

1. **Formulate Hypotheses**:
   - **Null Hypothesis (H0)**: There is no significant difference between the means of the two groups.
   - **Alternative Hypothesis (H1)**: There is a significant difference between the means of the two groups.
2. **Select the Appropriate Type**:
   - **One-Sample t-test**: Compare the mean of a single sample to a known value.
   - **Independent Samples t-test**: Compare the means of two independent groups.

- **Paired Samples t-test**: Compare the means of two related groups.

3. **Collect Data**:
   - Gather data from your two groups of interest.
4. **Calculate the t-statistic**:
   - Calculate the t-statistic using the formula appropriate for the type of t-test you are conducting.
5. **Determine the Degrees of Freedom**:
   - Calculate the degrees of freedom based on the sample sizes of the two groups.
6. **Consult the t-distribution Table or Software**:
   - Look up the critical t-value for your chosen significance level and degrees of freedom.
7. **Compare the Calculated t-value to the Critical t-value**:
   - If the calculated t-value is greater than the critical t-value, you reject the null hypothesis.
8. **Interpret the Results**:
   - Based on the comparison, determine whether there is a statistically significant difference between the means of the two groups.
9. **Report the Findings**:
   - Present the results, including the t-value, degrees of freedom, and p-value, to communicate the statistical significance of your findings.

Chi squared (χ^2) Test

The **Chi-squared test** is a statistical hypothesis test used to analyse contingency tables when dealing with large sample sizes. This test is primarily employed to determine whether two categorical variables are independent of each other.

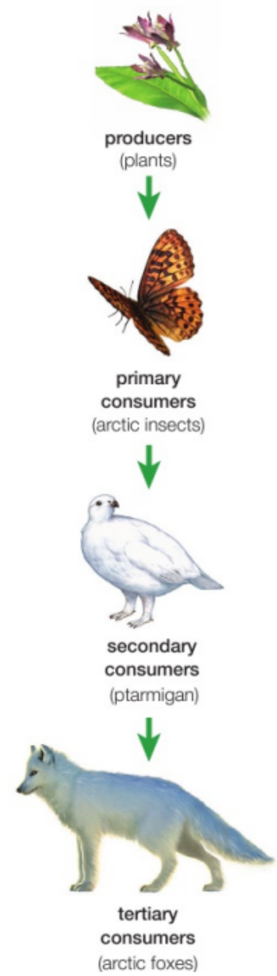
1. **Understand the Test**:
   - The Chi-squared test assesses the relationship between two categorical variables.
   - It compares observed frequencies with expected frequencies to determine if there is a significant association between the variables.
2. **Formulate Hypotheses**:
   - **Null Hypothesis (H0)**: There is no significant relationship between the variables.
   - **Alternative Hypothesis (H1)**: There is a significant relationship between the variables.
3. **Collect Data**:
   - Gather data related to the categorical variables you want to analyse.

4. **Calculate the Test Statistic**:
    ◦ Compute the Chi-squared statistic based on the observed and expected frequencies of the variables.
5. **Determine Degrees of Freedom**:
    ◦ Calculate the degrees of freedom to interpret the Chi-squared statistic correctly.
6. **Consult Critical Values Table**:
    ◦ Refer to a Chi-squared critical values table to find the critical value for your chosen significance level and degrees of freedom.
7. **Compare Calculated Statistic with Critical Value**:
    ◦ Compare the calculated Chi-squared statistic with the critical value.
    ◦ If the calculated value is greater than the critical value, you reject the null hypothesis.
8. **Interpret Results**:
    ◦ Based on the comparison, determine whether there is a statistically significant relationship between the categorical variables.
9. **Report Findings**:
    ◦ Present the results, including the Chi-squared statistic, degrees of freedom, and p-value, to communicate the statistical significance of your analysis.

TRANSFERS BETWEEN TROPHIC LEVELS

Food chains

Food chains show the direction of energy transfer from one trophic level to the other. Most food chains have 5 trophic levels, and rarely 6.



Producers

- They make food from sunlight by the process of photosynthesis.
- The plants and algae make glucose from CO_2 and H_2O.

Primary Consumers

- They feed off the producers
- They are herbivores
- Energy used for respiration and tissue formation

Secondary Consumers

- They feed off herbivores
- They are carnivores (unless they are also primary consumers in another food chain)
- They use the energy from plants for respiration and tissue formation

Tertiary Consumers

- They feed off other carnivores
- They are sometimes the apex predator in that area unless a quaternary consumer is there.

- They use the energy from secondary consumers for respiration and tissue formation.

Decomposers

- They are the final trophic level as all dead organisms and faeces are fed on by decomposers.
- They are mostly microorganisms that breakdown the remains and pass the nutrients back to the soil where they can be taken into plants to be used again
- They aren't usually included in food chains.

Food Webs

Since food chains don't show the relationships between organisms and all other organisms it eats or is eaten by, a food web is used **to show the connections between different food chains**.

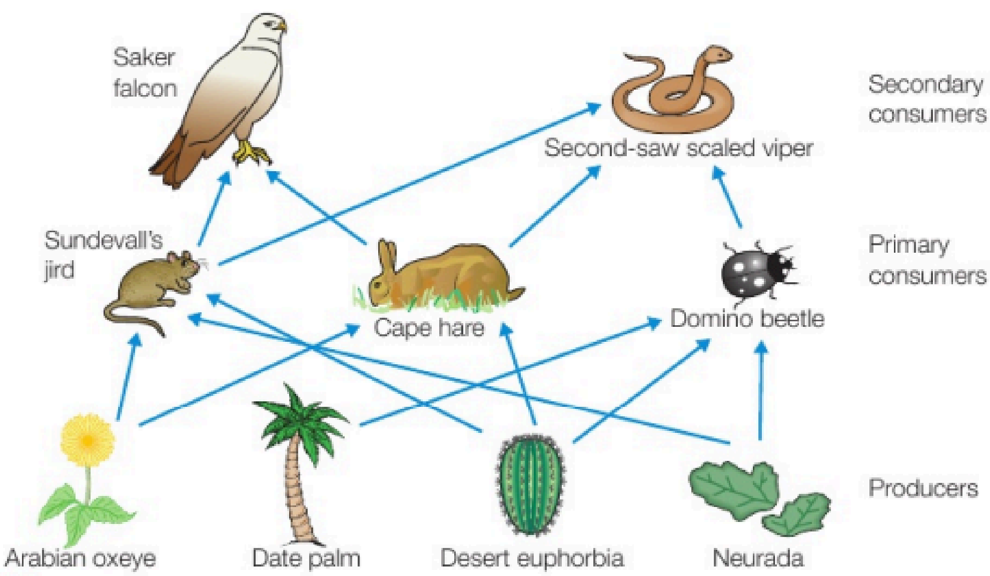


fig B A simplified food web from the United Arab Emirates. In a food web like this, if the numbers of one of the organisms drop, it will not be a disaster for the other organisms as they all have more than one food source.

Organisms part of a single food chain are very vulnerable and can go extinct if the food chain is disrupted. For example: if the bamboo plant decreases, pandas are threatened.

However, if the organism is part of a food web, the loss of one type of food can be made up for by other food sources. Still, there will be a change in balance, but not as severe as in single food chains.

Ecological Pyramids

Pyramids Of numbers

- A pyramid of numbers is a graphical representation that shows the numerical abundance of organisms at different levels in a food chain.
- It helps in understanding the feeding relationships and energy flow within an ecosystem.
- The base of the pyramid typically represents the organism with the highest number, while the top represents the organism with the lowest number.
- By comparing trophic levels, pyramids of numbers provide insights into the structure and dynamics of ecosystems.

Pyramids Of Biomass

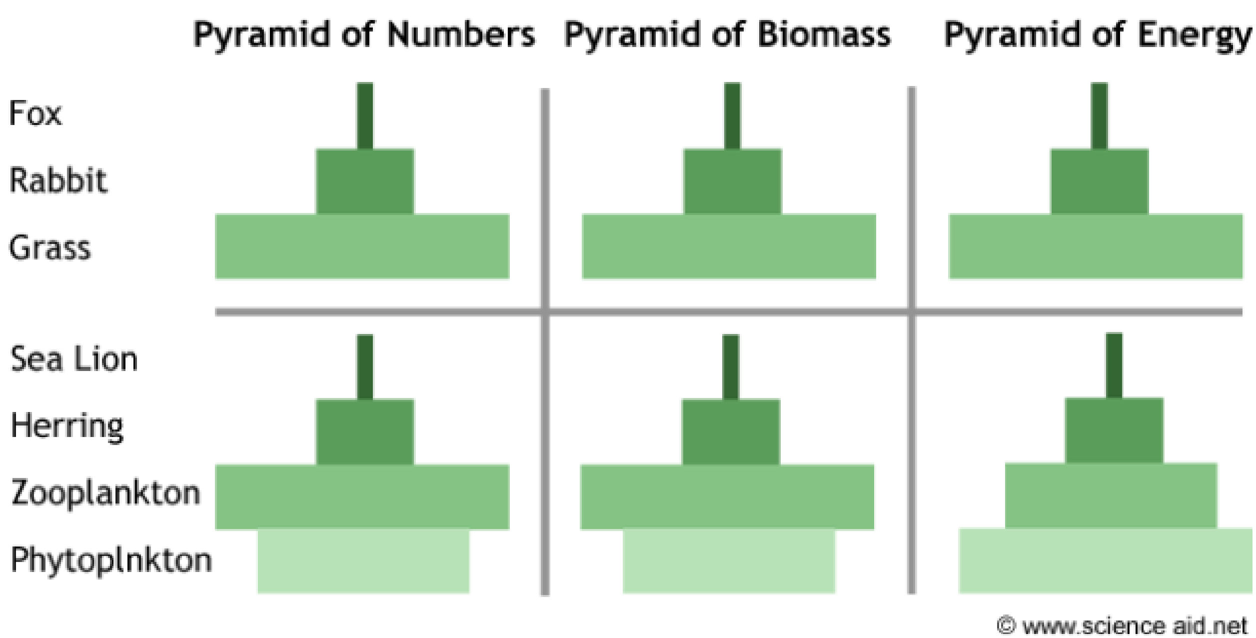
Pyramids of biomass are graphical representations of the total living biomass or organic matter present at different trophic levels in an ecosystem. They show the relationship between the biomass (the amount of living organisms) at each trophic level.

- A biomass pyramid collects the mass of each organism in a given habitat and organises them based on their trophic level through a pyramid model.
- They are always perfectly shaped, and if not, it may indicate an unhealthy or endangered ecosystem.
- The pyramid of biomass is a graphical portrayal of biomass of different trophic levels present in a unit area of an ecosystem.
- They represent the mass of organisms at each trophic level, showing the flow of energy through the ecosystem.

Pyramids Of Energy

Pyramids of energy are graphical representations that show the flow of energy at each trophic level in an ecosystem. They illustrate the amount of energy available at each level and how it decreases as you move up the food chain. Here are some key points about pyramids of energy:

- An energy pyramid is also known as a trophic or ecological pyramid.
- It demonstrates the transfer of energy in an ecosystem.
- The higher an organism is on the pyramid, the lower the available energy.
- Examples of ecosystems where energy pyramids are applicable include forest ecosystems.



Losses In A Food Chain

The energy transfer between trophic levels isn't completely efficient: most of the energy is lost.

Producers

- Most of the light from the sun passes through, gets reflected off the leaves, or cannot be used by the plant(wavelengths other than the visible light).
- Light falls on non-photosynthetic parts of the plant
- Energy lost during respiration.

Consumers

- Not all parts of the producers are eaten by the consumers (eg. roots)
- Energy is used for respirations
- Some parts of the producers aren't digested.
- Some energy is lost through excretory products.

Therefore, only about 10% of energy is passed from one trophic level to the other.

Sun
↓ 2 000 000 kJ

1. Plant 20 000kJ
   ↓ 10% to Primary Consumer
2. Primary Consumer 200kJ
   ↓ 10% to Secondary Consumer
3. Secondary Consumer 2kJ
   ↓ 10% to Tertiary Consumer
4. Tertiary Consumer 0.02kJ (20J)
   ↓ 10% to Quaternary Consumer
5. Quaternary Consumer 0.0002kJ (0.2J)

This is what limits the number of trophic levels in a food chain. A quaternary consumer can only be supported where the energy reaching the producers is very high.

NET PRIMARY PRODUCTIVITY

NPP AND GPP

Gross Primary Productivity (GPP) is the total amount of energy, mainly from sunlight, that plants in an ecosystem capture and convert into food. It is the total amount of carbon compounds produced by photosynthesis of plants in an ecosystem. GPP is a key measure of the productivity and energy flow within an ecosystem.

GPP can be measured in g m^{-1} $year^{-1}$, or g C m^{-1} $year^{-1}$, or kJ m^{-1} $year^{-1}$

Net Primary Production (NPP) is the difference between the energy fixed by autotrophs and their respiration (NPP = GPP - Respiration). It is commonly defined as the amount of carbon retained in an ecosystem, which represents an increase in biomass. NPP can also be described as the rate of accumulation of biomass or energy in an ecosystem. It is a crucial measure of the productivity and health of an ecosystem.

NPP depends on abiotic factors that affect plant growth.

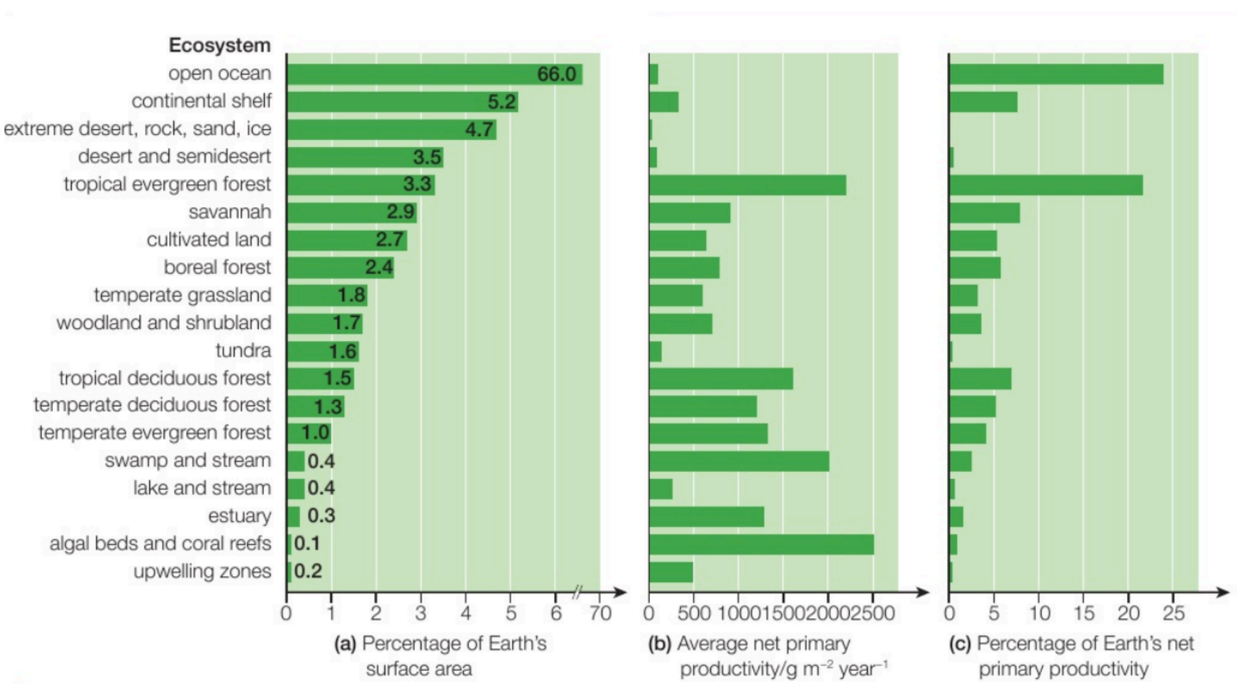


fig C Different ecosystems contribute different amounts to the NPP of the Earth.

Energy Transfers To Higher Trophic Levels

Calculating energy transfers from one trophic level to another becomes complicated when decomposers are considered.

THE CARBON CYCLE

Overview of the Carbon Cycle

The carbon cycle is a crucial biogeochemical cycle that describes the movement of carbon through the Earth's atmosphere, hydrosphere, lithosphere, and biosphere. Carbon is a fundamental element of life, found in all living organisms, and it also plays a significant role in regulating Earth’s climate.

Key Processes in the Carbon Cycle

1. **Photosynthesis**: Plants, algae, and some bacteria absorb carbon dioxide (CO_2) from the atmosphere and convert it into organic compounds (such as glucose) during photosynthesis. This process is the primary method by which carbon enters the biosphere.
2. **Respiration**: All living organisms (plants, animals, and microorganisms) release carbon dioxide back into the atmosphere through respiration, a process that involves the breakdown of organic molecules to release energy.
3. **Decomposition**: When organisms die, decomposers (bacteria and fungi) break down the organic material, releasing carbon dioxide back into the atmosphere or soil.
4. **Combustion**: The burning of organic material, such as fossil fuels (coal, oil, natural gas) and biomass, releases stored carbon as carbon dioxide into the atmosphere.
5. **Ocean Uptake and Release**: The oceans absorb carbon dioxide from the atmosphere, where it can dissolve in water or be used by marine organisms for photosynthesis. Carbon can also be released from the oceans back into the atmosphere, depending on temperature and other factors.

6. **Sedimentation and Fossilization**: Over long geological periods, some organic material is buried and converted into fossil fuels. Additionally, carbon can be stored in sedimentary rocks, such as limestone, which is formed from the accumulation of calcium carbonate from marine organisms.

Carbon Sinks

Carbon sinks are natural systems that absorb more carbon than they release, thus helping to reduce the amount of CO_2 in the atmosphere. They play a critical role in mitigating the impact of human activities on the carbon cycle.

1. **Forests**: Trees and other vegetation act as significant carbon sinks by absorbing CO_2 during photosynthesis and storing it as biomass (wood, leaves, roots).
2. **Oceans**: The world's oceans are the largest carbon sink, absorbing about 25-30% of the CO_2 emitted by human activities. CO_2 dissolves in seawater and can be used by marine organisms or stored in the deep ocean for long periods.
3. **Soil**: Soil contains organic matter, including decaying plants and animals, which stores carbon. Healthy, undisturbed soils can sequester significant amounts of carbon.
4. **Sedimentary Rocks**: Over geological timescales, carbon is stored in sedimentary rocks, such as limestone, which is formed from the skeletal remains of marine organisms.

Human Influence on the Carbon Cycle

Human activities have significantly altered the carbon cycle, particularly through the increased release of carbon dioxide and other greenhouse gases. Key influences include:

1. **Burning of Fossil Fuels**: The combustion of coal, oil, and natural gas for energy has dramatically increased the concentration of CO_2 in the atmosphere. This process releases carbon that has been stored underground for millions of years, disrupting the natural balance of the carbon cycle.
2. **Deforestation**: The clearing of forests for agriculture, urban development, and other purposes reduces the number of trees that can absorb CO_2 from the

atmosphere. Additionally, burning or decomposing cleared vegetation releases stored carbon back into the atmosphere.

3. **Agriculture**: Agricultural practices, including livestock production, rice paddies, and the use of synthetic fertilizers, contribute to the release of methane (CH_4) and nitrous oxide (N_2O), both potent greenhouse gases. Tilling the soil also releases stored carbon.
4. **Land Use Changes**: Urbanization and other land use changes reduce the area of natural carbon sinks, such as forests and wetlands, and replace them with less effective carbon stores, like urban landscapes.
5. **Ocean Acidification**: Increased CO_2 levels in the atmosphere lead to higher CO_2 absorption by the oceans, which results in ocean acidification. This process decreases the ability of marine organisms, such as corals and shellfish, to form calcium carbonate shells, affecting the marine carbon sink.

Consequences of Human Influence

- **Climate Change**: The increased levels of CO_2 and other greenhouse gases in the atmosphere trap more heat, leading to global warming and associated climate changes, such as more extreme weather events, rising sea levels, and changing ecosystems.
- **Disruption of Ecosystems**: Changes in the carbon cycle can disrupt ecosystems, affecting biodiversity and the availability of natural resources.
- **Oceanic Changes**: Ocean acidification, as a result of increased CO_2 absorption, threatens marine life and the overall health of ocean ecosystems, which can lead to a reduction in the effectiveness of the ocean as a carbon sink.

Mitigating Human Impact

- **Reducing Fossil Fuel Use**: Transitioning to renewable energy sources, such as solar, wind, and hydropower, can reduce CO_2 emissions.
- **Reforestation and Afforestation**: Planting new forests and restoring degraded ones can increase the number of carbon sinks, helping to absorb more CO_2 from the atmosphere.

- **Sustainable Agriculture**: Implementing sustainable farming practices can reduce the release of greenhouse gases and increase carbon sequestration in soils.
- **Carbon Capture and Storage (CCS)**: Technologies that capture CO_2 emissions from power plants and industrial processes and store them underground can help reduce atmospheric CO_2 levels.

Summary

The carbon cycle is a complex system of carbon exchanges between the atmosphere, oceans, land, and living organisms. Human activities, particularly the burning of fossil fuels and deforestation, have significantly disrupted this cycle, leading to increased atmospheric CO_2 levels and contributing to climate change. Understanding and managing carbon sinks, alongside reducing emissions, are crucial for mitigating the impacts of human influence on the carbon cycle and ensuring the sustainability of Earth's ecosystems.

TOPIC 6